9.6 Right Lane Closed – Short and Long Duration

Purpose:

This layout shows the typical setup for closure of the right travel lane on a multilane divided highway.

Standard:

- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab and a second Right Lane Closed Ahead C-130-R sign shall be placed in advance of the work.

Guidance:

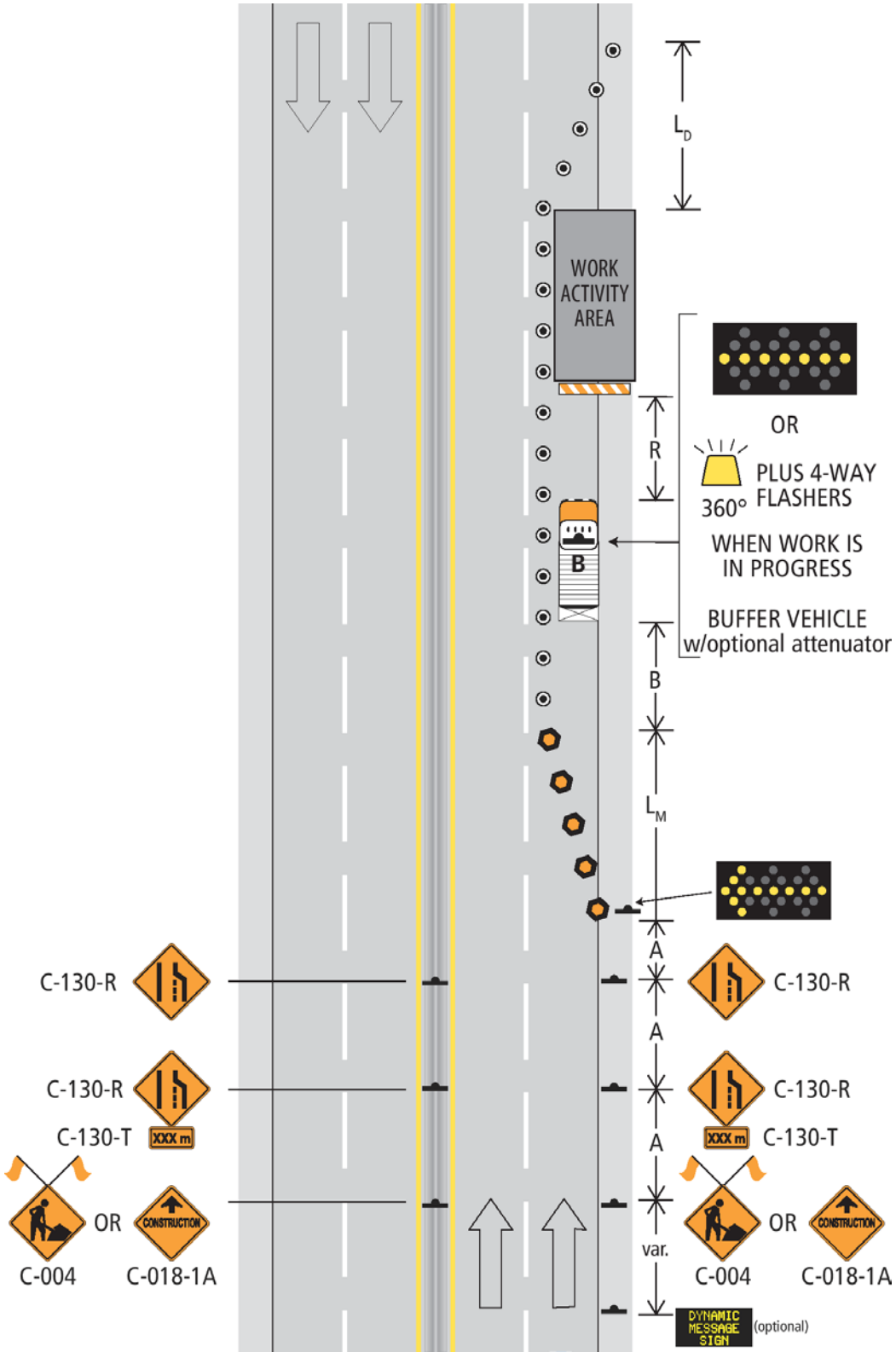
- Signage should be installed along the divided median. If the median space is limited, smaller sized signs may be used.
- Where shoulders are wide and may be mistaken for travel lanes, shoulder tapers should be considered and installed in accordance with *Table A – Taper Lengths*.

Options:

- If the regular posted speed limit is ≤ 60 km/h:
  - The upstream Right Lane Closed C-130-R sign with distance tab may be omitted and the Crew Working Ahead C-004 sign or Construction Ahead C-018-1A sign moved downstream by Table B Distance A.
  - Tubular markers may be used for leading tapers instead of drums.
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 9.6: Right Lane Closed – Short and Long Duration



9.7 Left Lane Closed – Short and Long Duration

Purpose:

This layout shows the typical setup for closure of the left travel lane on a multilane divided highway.

Standard:

- A Left Lane Closed Ahead C-130-L sign with a C-130-T distance tab and a second Left Lane Closed Ahead C-130-L sign shall be placed in advance of the work.
- Where the posted speed limit is ≥ 70 km/h, a flashing arrow board (FAB) is required.

Guidance:

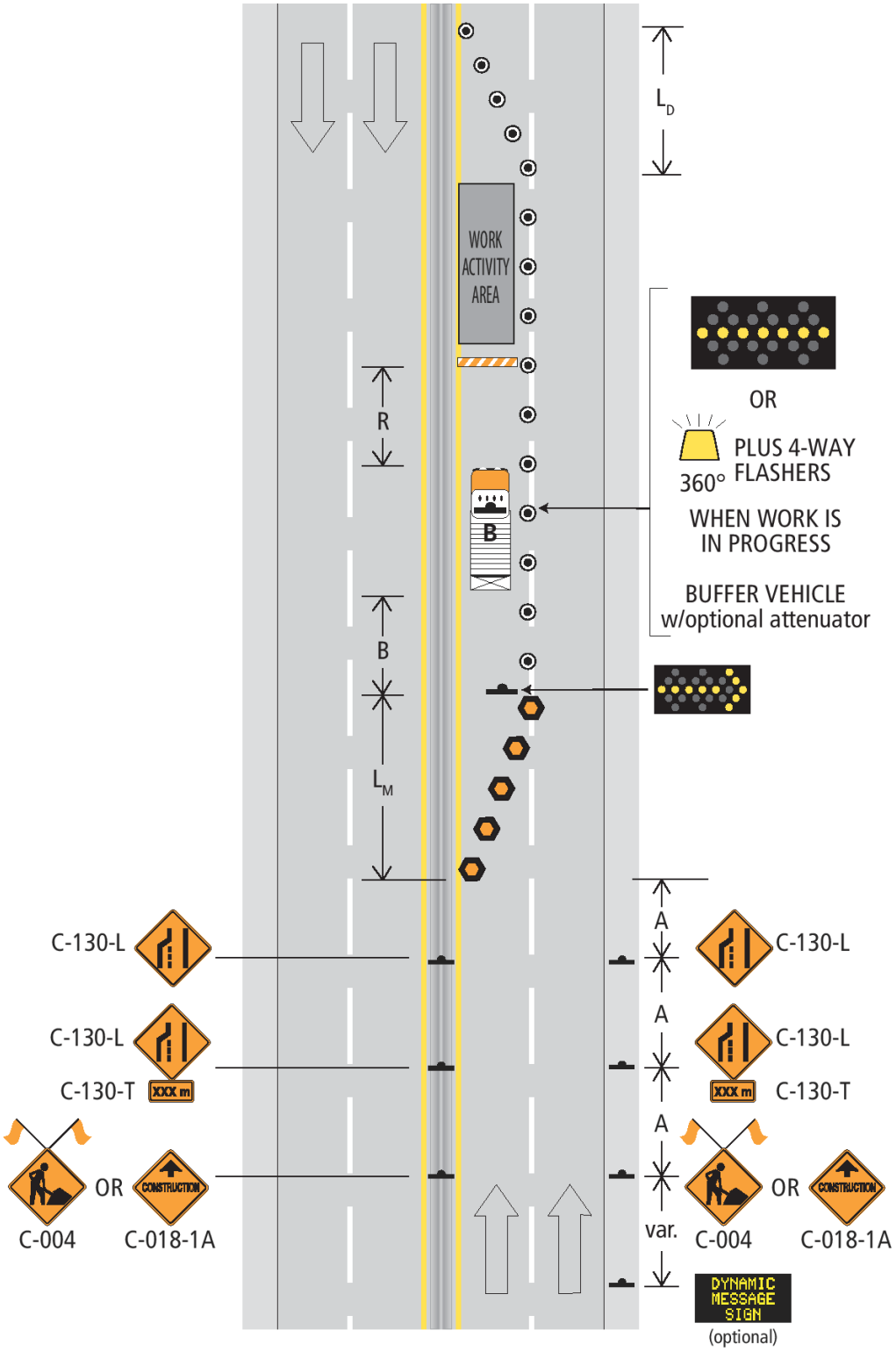
- Signage should be installed along the divided median. If the median space is limited, smaller sized signs may be used.

Options:

- If the regular posted speed limit is ≤ 60 km/h:
  - The upstream Left Lane Closed C-130-L sign may be omitted and the Crew Working Ahead C-004 sign or Construction Ahead C-018-1A sign moved downstream by Table B Distance A.
  - Tubular markers may be used for leading tapers instead of drums.
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 9.7: Left Lane Closed – Short and Long Duration



9.8 Centre Lane Closure (≤ 60 km/h) – Short and Long Duration

Purpose:

This layout shows the typical setup for a single lane closure of the centre lane on a low-speed multilane roadway.

Standard:

- A Centre Lane Closed Ahead C-030-1A sign with a C-130-T distance tab and a Centre Lane Closed C-030-2 sign are required in advance of the work.
- A buffer space shall be incorporated into the layout.

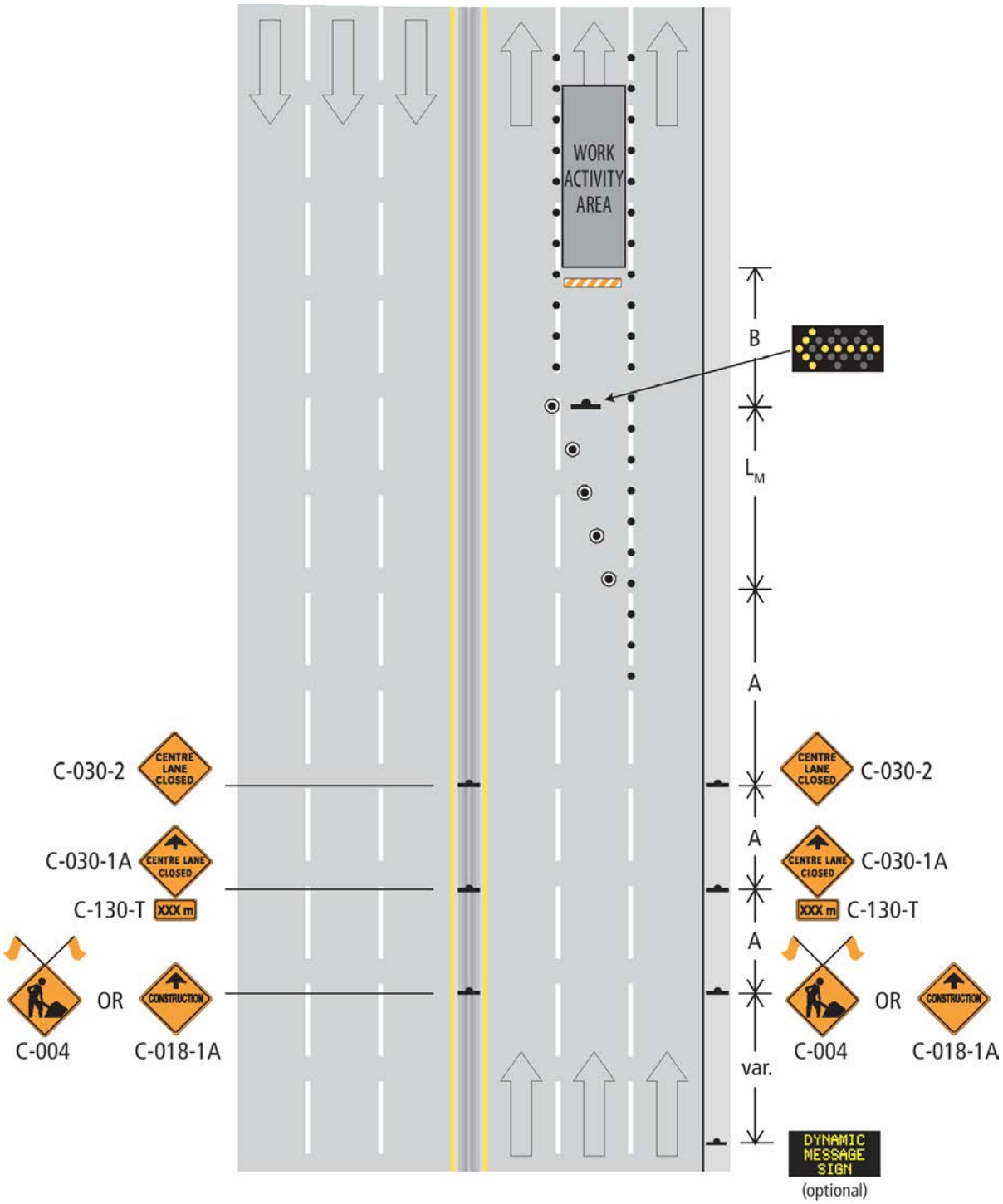
Guidance:

- The spacing of channelizing devices may be reduced to prevent traffic from entering the work area.
- Provided that traffic volumes are sufficiently low, it may be necessary to close two lanes to maintain adequate space for worker safety.
- If space is available, a short, single row of channelizing devices should be installed in advance of the traffic split to keep vehicles in their lanes.

Options:

- Signage may be installed along the divided median. If the median space is limited, smaller sized signs may be used.
- Interior lane traffic may be directed to either the left or the right lane by using a flashing arrow board (FAB) with the arrow pointing in the direction of the merge taper.

Figure 9.8: Centre Lane Closure (≤ 60 km/h) – Short and Long Duration



9.9 Centre Lane Closure (≥ 70 km/h) – Short and Long Duration

Purpose:

This layout shows the typical setup for a single lane closure of the centre lane on a high-speed multilane roadway.

Standard:

- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab and a second Right Lane Closed Ahead C-130-R sign shall be placed in advance of the work.
- A Centre Lane Closed C-030-2 sign is required in advance of the lane shift.
- If signs are not installed on the divided median, a dynamic message sign (DMS) is required with the message “CENTRE LANE CLOSED”.
- Two flashing arrow boards (FABs) shall be used—one in each taper—as illustrated in the diagram.

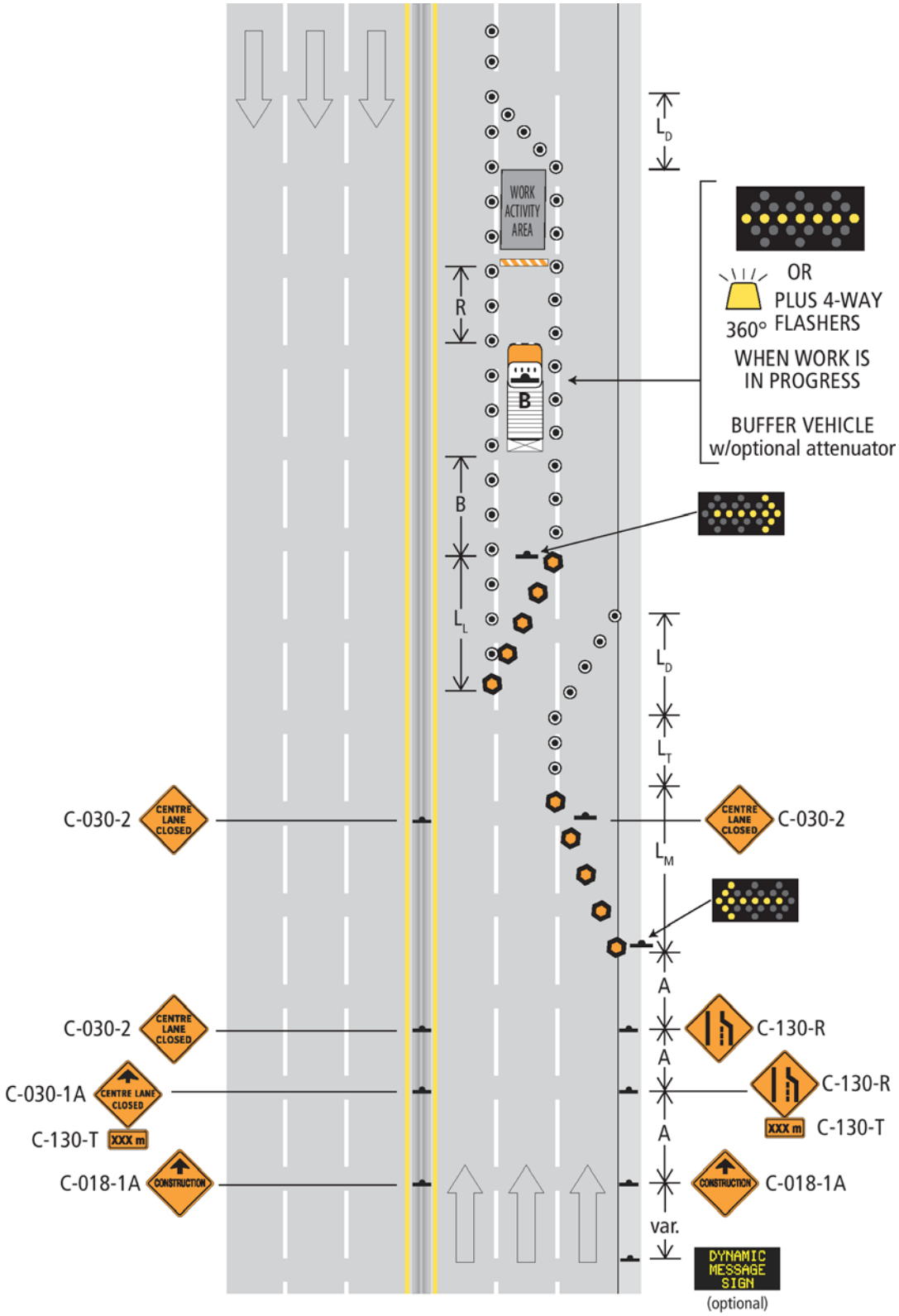
Guidance:

- If the alignment is such that the two flashing arrow boards (FABs) create confusion, the minimum tangent length (L_T distance) between the end of the merging taper and beginning of the shift taper should be extended so that road users can focus on one flashing arrow board (FAB) at a time.
- The spacing of channelizing devices may be reduced to prevent traffic from entering the work area.
- A double lane closure (leaving only one lane open) may be implemented if traffic volumes are sufficiently low, using a flashing arrow board (FAB) in the tapers for each closed lane.
- Where shoulders are wide and may be mistaken for travel lanes, shoulder tapers should be considered and installed in accordance with *Table A – Taper Lengths*.

Options:

- Interior lane traffic may be directed to either the left or the right lane by using a flashing arrow board (FAB) with the arrow pointing in the direction of the merge taper.

Figure 9.9: Centre Lane Closure (≥ 70 km/h) – Short and Long Duration



9.10.1 Double Right Lane Closure – Short and Long Duration

Purpose:

This layout shows the typical setup for the closure of two right lanes in a single travel direction on a multilane divided highway where at least one lane in the same direction of travel remains available to traffic.

For closures of multiple lanes, it is important that drivers have to navigate only one lane closure at a time (i.e., each closed lane will have a separate taper and merge setup). In other words, double lane closures should never require drivers to make two lane merges at one location.

Standard:

- Each lane shall be closed one at a time as shown in the layout.
- A separate flashing arrow board (FAB) shall be used for each closed lane.

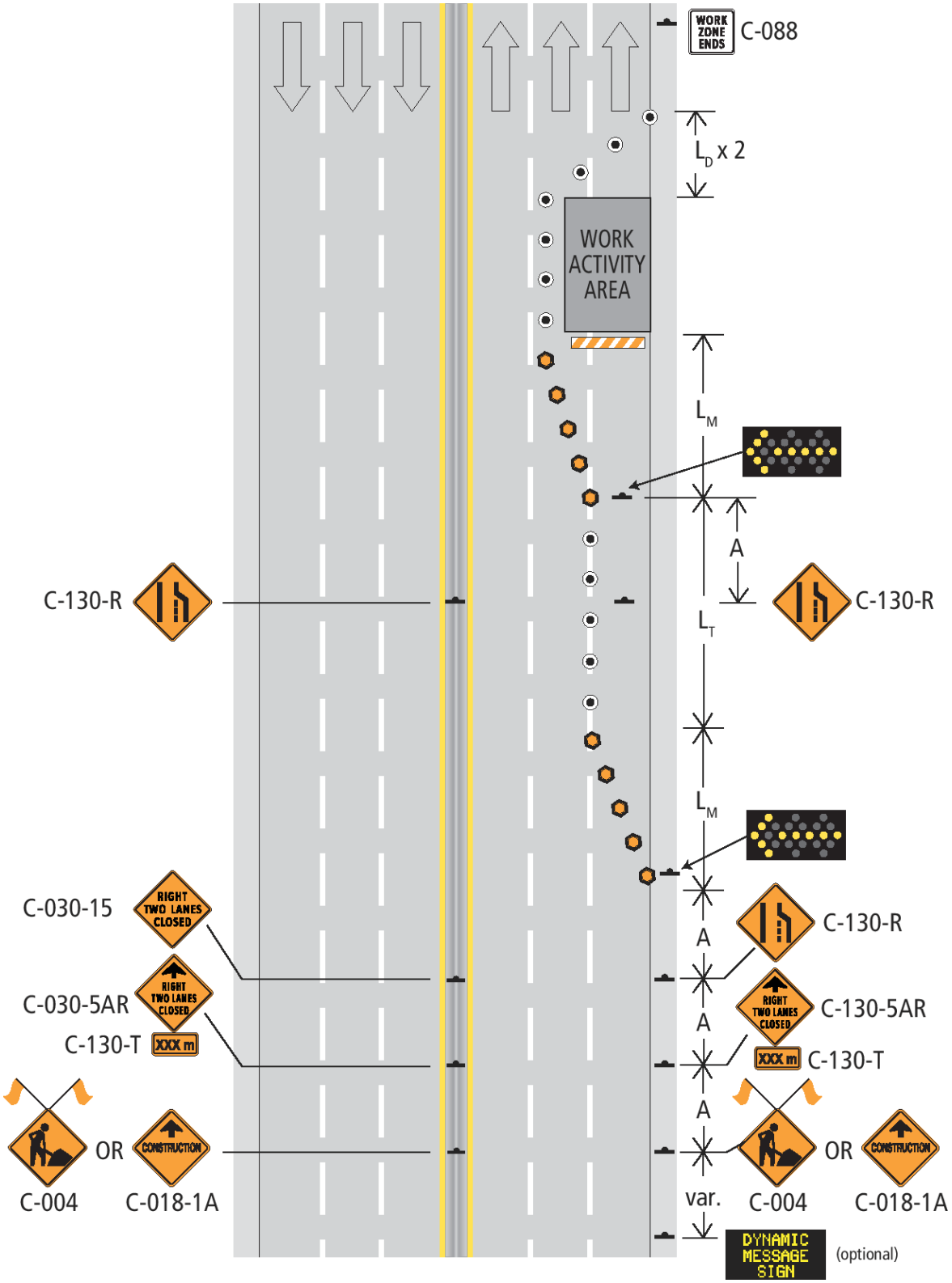
Guidance:

- Signage should be installed along the divided median. If the median space is limited, smaller sized signs may be used.
- Where shoulders are wide and may be mistaken for travel lanes, shoulder tapers should be considered and installed in accordance with *Table A – Taper Lengths*.

Options:

- For long-duration work, a dynamic message sign (DMS) may be used in advance of the Construction Ahead C-018-1A sign.
- L_T and the positioning of the second FAB may be adjusted if the alignment creates confusion about which lane closure is being indicated by the second FAB.

Figure 9.10.1: Double Right Lane Closure – Short and Long Duration



9.10.2 Double Left Lane Closure – Short and Long Duration

Purpose:

This layout shows the typical setup for the closure of two left lanes in a single travel direction on a multilane divided highway where at least one lane in the same direction of travel remains available to traffic.

For closures of multiple lanes, it is important that drivers have to navigate only one lane closure at a time (i.e., each closed lane will have a separate taper and merge setup). In other words, double lane closures should never require drivers to make two lane merges at one location.

Standard:

- Each lane shall be closed one at a time as shown in the layout.
- A separate flashing arrow board (FAB) shall be used for each closed lane.

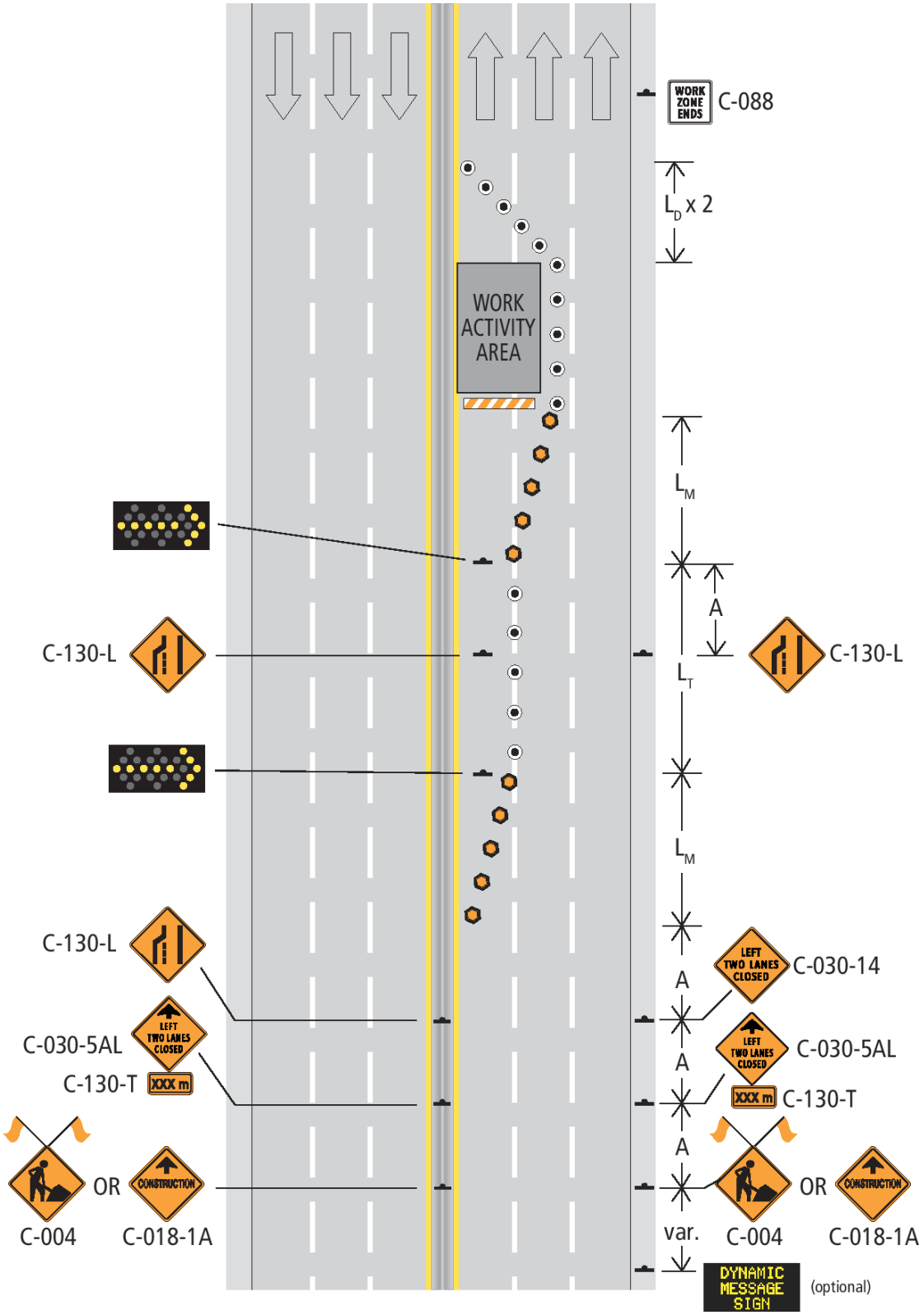
Guidance:

- Signage should be installed along the divided median. If the median space is limited, smaller sized signs may be used.
- Where shoulders are wide and may be mistaken for travel lanes, shoulder tapers should be considered and installed in accordance with *Table A – Taper Lengths*.

Options:

- For long-duration work, a dynamic message sign (DMS) may be used in advance of the Construction Ahead C-018-1A sign.
- L_T and the positioning of the second FAB may be adjusted if the alignment creates confusion about which lane closure is being indicated by the second FAB.

Figure 9.10.2: Double Left Lane Closure – Short and Long Duration



9.11 Median Crossover – Short and Long Duration

Purpose:

This layout shows the typical setup for the closure of all lanes in an entire direction of travel on a multilane roadway. Traffic from the closed lanes is re-routed across the median into the opposing travel lane, reducing the number of lanes available for opposing traffic.

This layout reduces road capacity in both directions of travel so consideration shall be given to the best time of day for implementing this type of closure.

Standard:

- A Lane Closed Ahead C-130-L/R sign will be required for the left/right lane closure in advance of the work.
- Channelizing devices or temporary barriers shall be used to separate opposing traffic.
- Two-Way Traffic Ahead C-132 signs shall be used when concrete median barrier is not used to define the new centreline.
- If median barrier is removed to enable the median crossover, the remaining exposed barrier ends shall be marked by an appropriate Hazard C-154 marker and a Type B flasher.
- If the ends of the barrier will be exposed for longer than 48 hours, suitable end treatment shall be applied as determined by the Road Authority.
- If signs are not installed on the divided median, a dynamic message sign (DMS) is required.

Guidance:

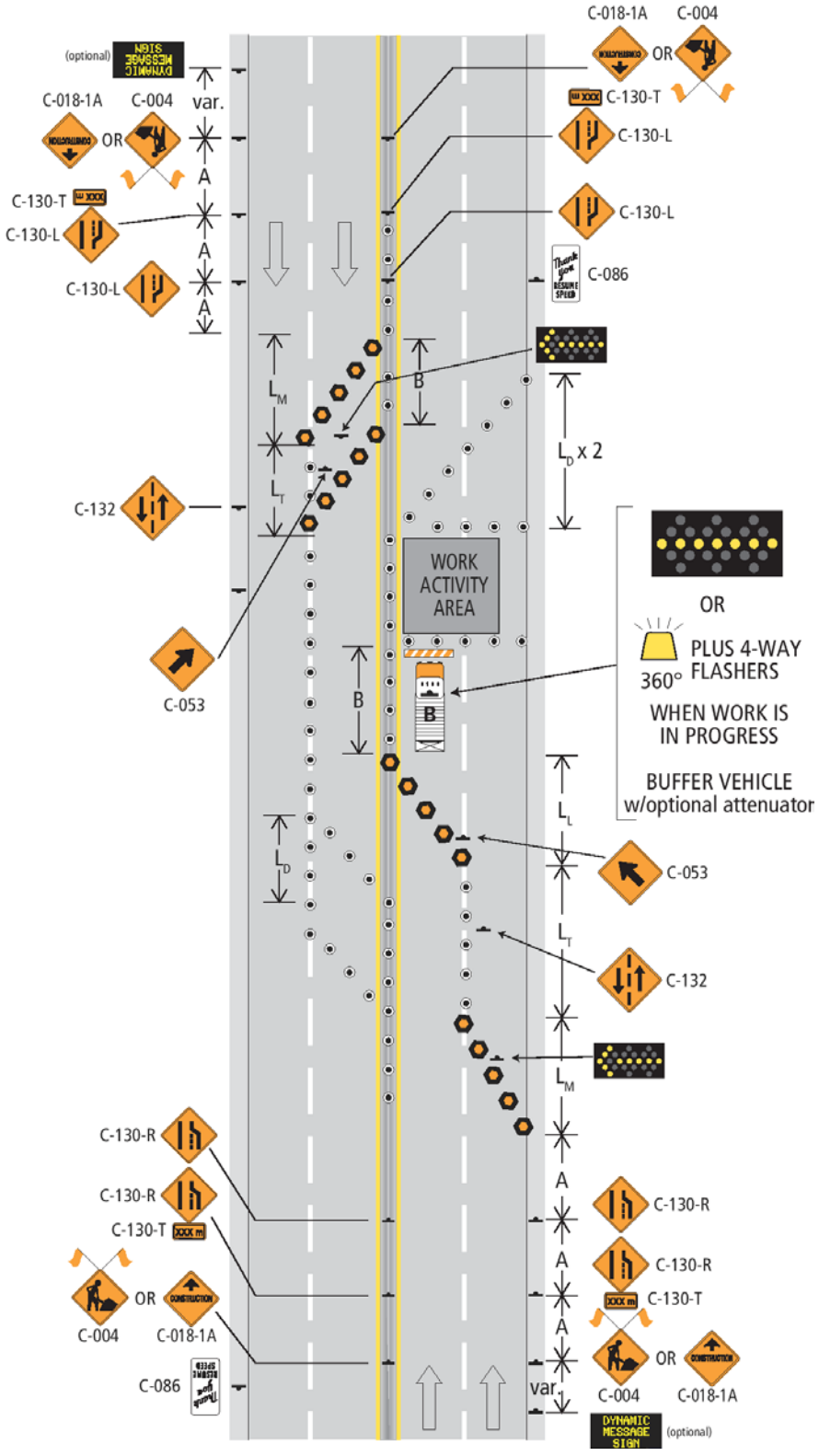
- Temporary crash attenuators should be used to protect the exposed ends of median barriers.
- The Barrier Removed C-069 sign should be used in advance of locations where the existing median barrier has been removed.
- When re-establishing normal traffic flow, remove the signs and devices for the diverted lanes first, and then remove the devices for the opposing lanes.
- A dynamic message sign (DMS) is recommended for night work over multiple night-time shifts.

Options:

- A custom Traffic Pattern Changed Zx-030 sign with an implementation plan on counterflow operations and transitions may be used as a supplement. See Zx-030 *Appendix B*.
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 9.11: Median Crossover – Short and Long Duration



9.12 Lane Closure at Open Exit Ramp – Short and Long Duration

Purpose:

This layout shows the typical setup where there is a lane closure in the vicinity of an exit ramp but the ramp itself remains open.

The presence of construction activity and traffic control devices associated with the lane closure may cause driver uncertainty about the status of the exit ramp so it is important to use appropriate delineation and signage to let drivers know in advance—and at the exit—that the exit is open.

Standard:

- The Exit Open C-204 sign with a C-130-T distance tab shall be used in advance of the open exit to identify the distance to the new exit point.
- An additional Exit Open C-204 sign shall be used in advance of the new exit point.

Guidance:

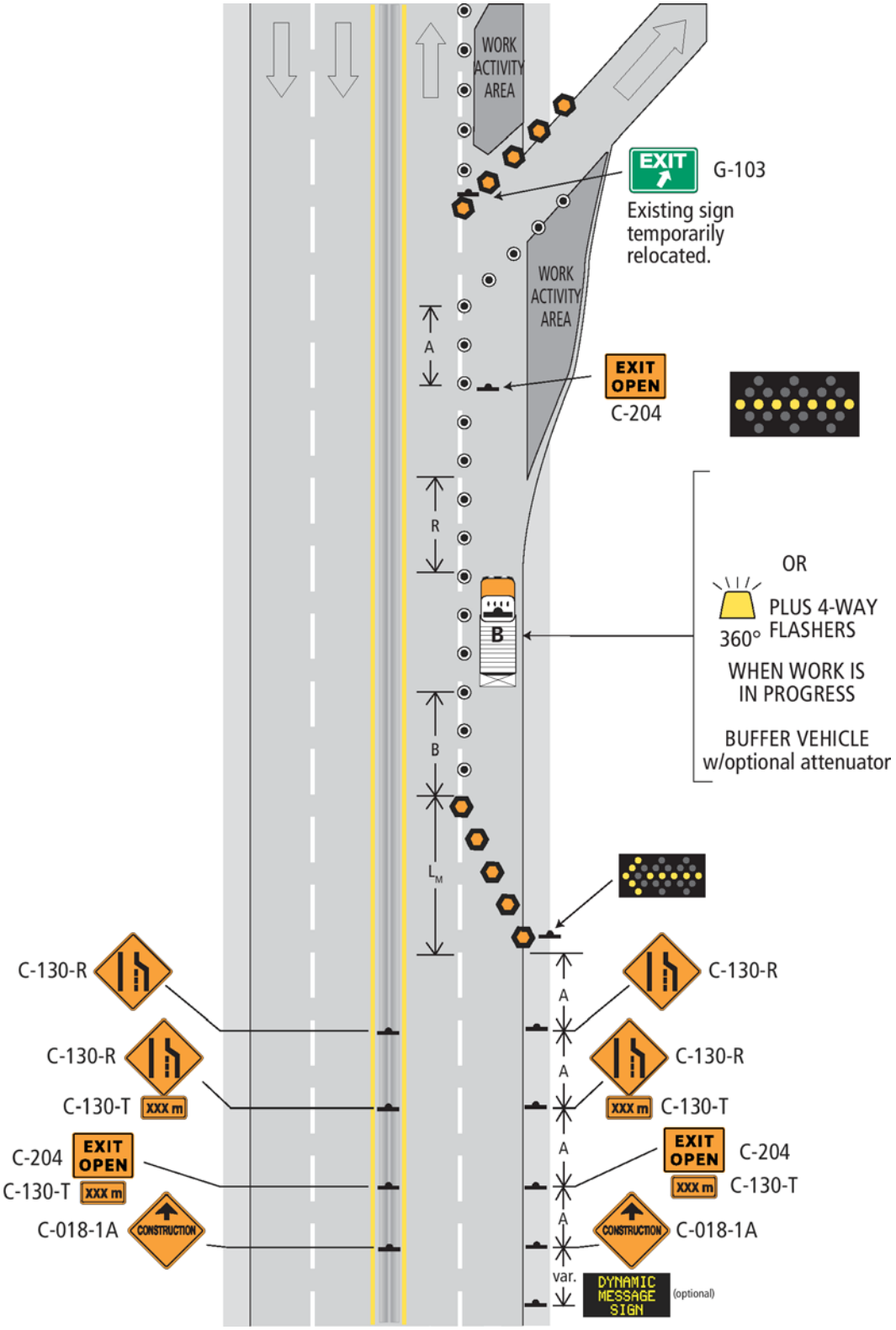
- Signage should be installed along the divided median. If the median space is limited, smaller sized signs may be used.
- The Exit G-103 sign should be used at the temporary exit to define the exit point, mounted over temporary channelizing devices in a manner that makes it highly visible to drivers.
- A dynamic message sign (DMS) is recommended for night work over multiple night-time shifts.
- Channelizing devices used to define the path to the exit ramp should be tapered so that they are consistent with the ramp approach.

Options:

- A dynamic message sign (DMS) may be used to advise drivers that the exit is open.
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 9.12: Lane Closure at Open Exit Ramp – Short and Long Duration



9.13 Lane Closure at Open Entrance Ramp – Yield Condition – Short and Long Duration

Purpose:

This layout shows the typical setup where there is a lane closure in the vicinity of an entrance ramp but the ramp remains open. Construction activity near the entrance ramp may necessitate changing how vehicles enter the highway.

This layout specifically illustrates a yield condition for the entrance ramp. A yield condition shall be implemented when sufficient length cannot be maintained for a merge condition.

Standard:

- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab and a second Right Lane Closed Ahead C-130-R sign are required in advance of the entrance ramp.
- A Merging Traffic Ahead C-136-R sign is required in advance of the entrance ramp to advise drivers that traffic is merging from the right in the new location.
- The entrance ramp requires a Crew Working Ahead C-004 sign or a Construction Ahead C-018-1A sign and a Yield R-002 sign to advise those entering the roadway that they are in a yield condition.

Guidance:

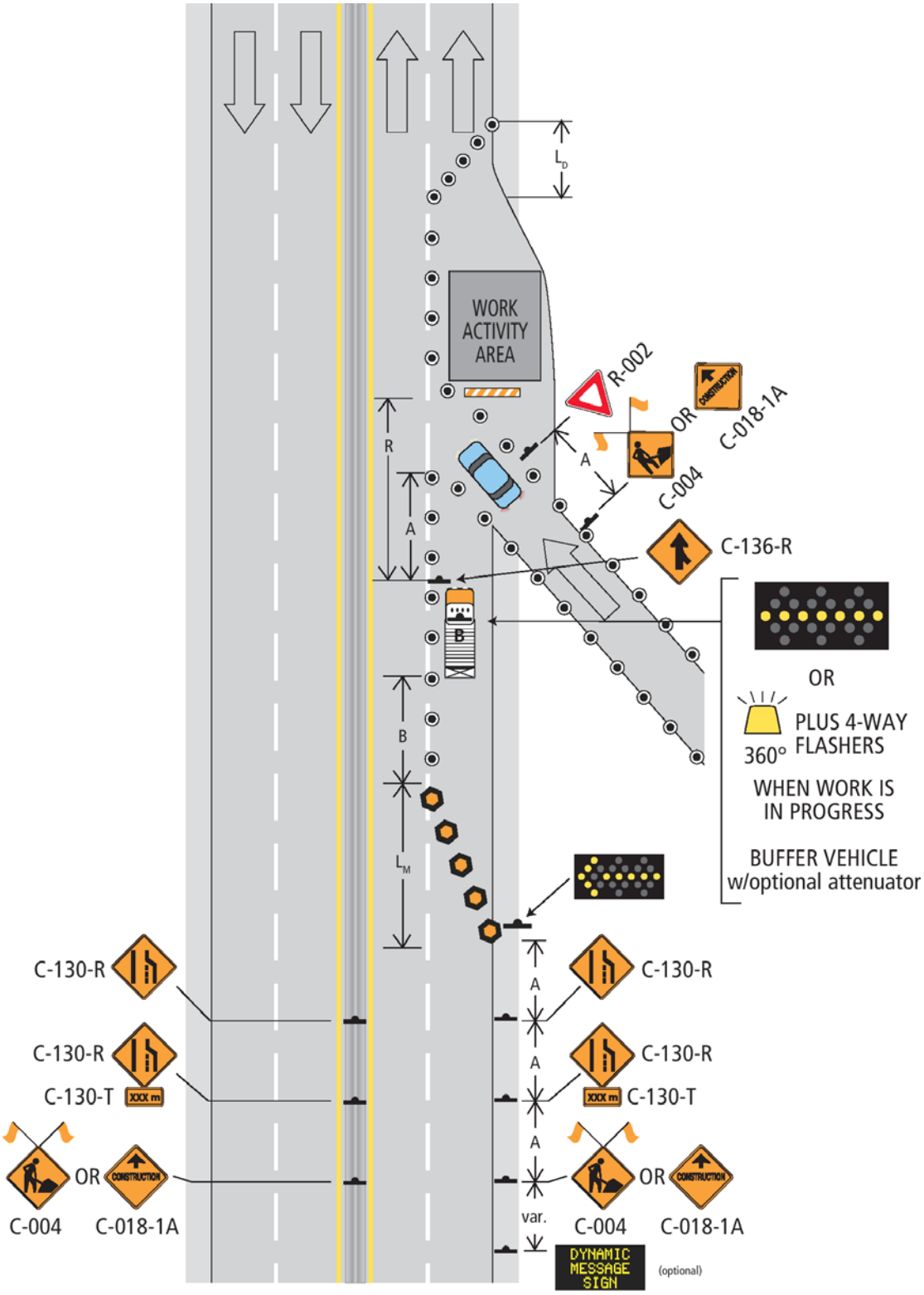
- Signage should be installed along the divided median. If the median space is limited, smaller sized signs may be used.
- The Yield R-002 sign should be positioned so that ramp traffic has adequate sight distance to select an acceptable gap in the traffic flow but not so far forward that drivers are tempted to stop in the major road traffic path.
- If insufficient gaps are available, consideration should be given to closing the ramp.
- Signs and devices should be positioned so they do not block major road drivers' view of the Merging Traffic C-136-R sign and the merging traffic.
- A dynamic message sign (DMS) is recommended for night work over multiple night-time shifts.

Options:

- If there is a construction speed limit in place, consider installing a confirmatory Maximum Speed Limit R-004 sign with a Construction Speed Limit C-080-T tab downstream of the on-ramp.
- To improve sight distance for the merging traffic, tubular markers between the ramp and the leading taper may be substituted with cones.
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 9.13: Lane Closure at Open Entrance Ramp – Yield Condition – Short and Long Duration



9.14 Lane Closure at Open Entrance Ramp – Merge Condition – Short and Long Duration

Purpose:

This layout shows the typical setup where there is a lane closure in the vicinity of an entrance ramp but the ramp remains open. Construction activity near the entrance ramp may necessitate changing how vehicles enter the highway.

This layout specifically illustrates a merge condition for the entrance ramp, which means that drivers entering the highway from the ramp are still able to use an acceleration lane to merge into traffic.

It is generally preferable to maintain ramp operations using a merge condition if an acceleration lane of sufficient length can be maintained.

Standard:

- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab and a second Right Lane Closed Ahead C-130-R sign are required in advance of the entrance ramp.
- A Merging Traffic Ahead C-136-R sign is required along the work zone tangent and in advance of the new entrance ramp location to advise drivers that traffic is merging from the right in the new location.
- An acceleration lane of sufficient length shall be provided for a merge condition.

Guidance:

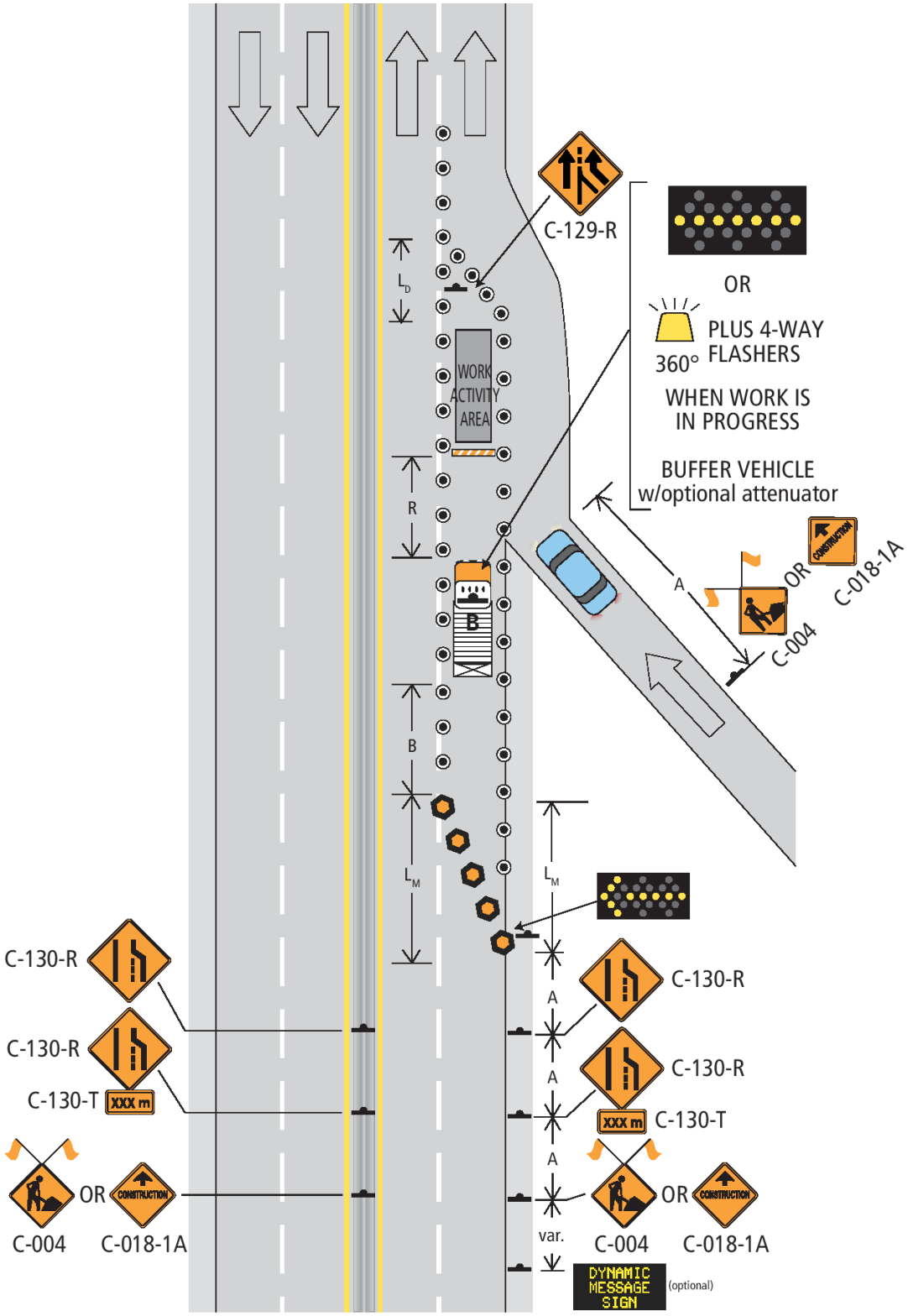
- Signage should be installed along the divided median. If the median space is limited, smaller sized signs may be used.
- Signs and devices should be positioned so that they do not block major road drivers' view of the Merging Traffic C-136-R sign and the merging traffic.
- A dynamic message sign (DMS) is recommended for night work over multiple night-time shifts.

Options:

- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 9.14: Lane Closure at Open Entrance Ramp – Merge Condition – Short and Long Duration



9.15 Temporary Closure of Exit Ramp – Short and Long Duration

Purpose:

This layout shows the typical setup where an exit ramp is closed.

The presence of construction activity and traffic control devices associated with the lane closure may cause driver uncertainty about the status of the exit ramp so it is important to use appropriate delineation and signage to let drivers know in advance—and at the exit—that the exit is closed.

It is important to clearly block off the entrance to the closed exit ramp and any deceleration lane.

Standard:

- The Exit Closed Ahead C-205-A sign shall be used in advance of a temporarily closed exit ramp.
- Drums shall be used to close off the deceleration lane and the exit ramp.
- An Exit Closed C-062 banner shall be attached to the existing Exit G-103 sign to indicate the closure.

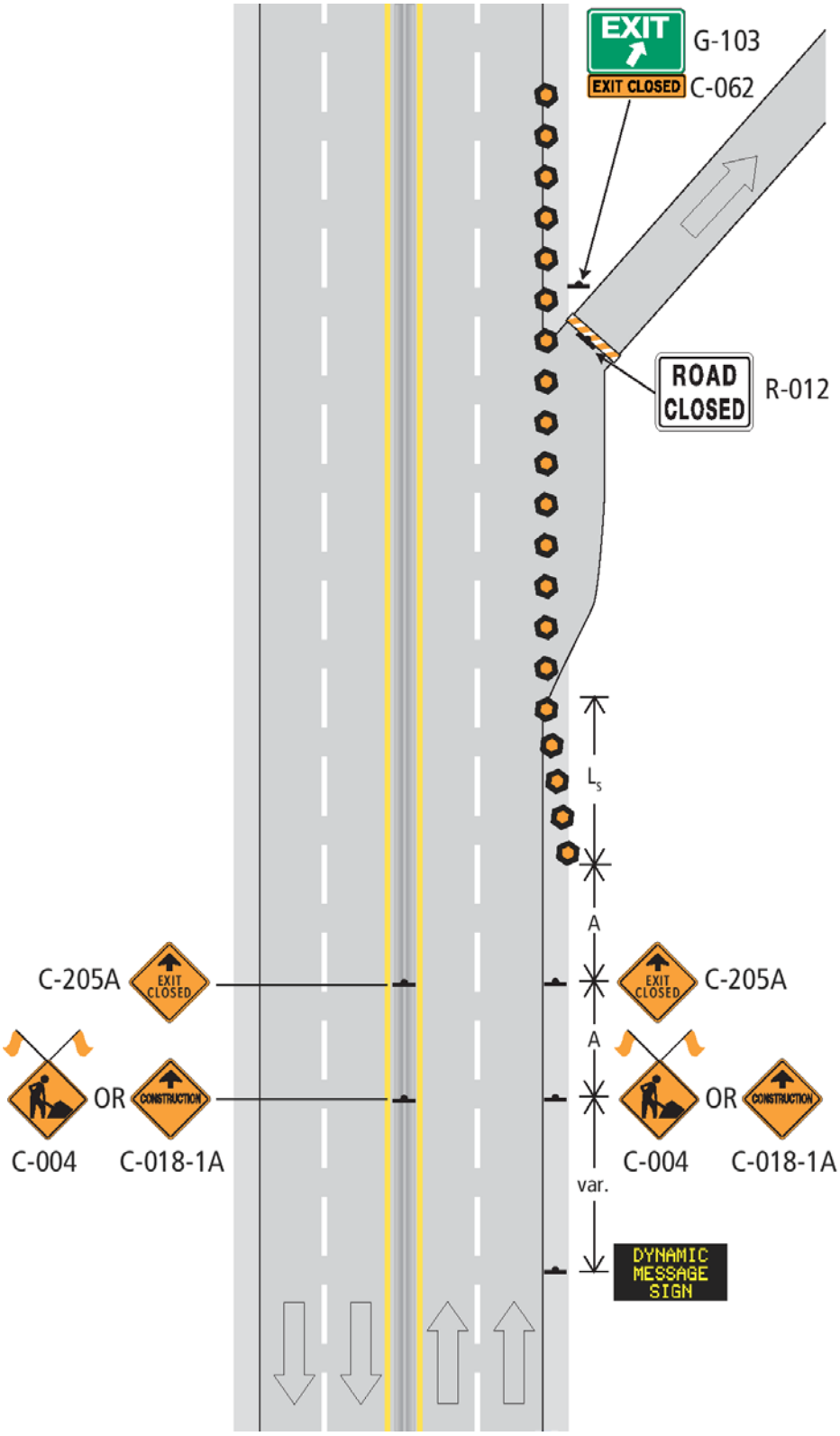
Guidance:

- Signage should be installed along the divided median. If the median space is limited, smaller sized signs may be used.
- Additional Exit Closed Ahead C-205-A signs may be required for higher-volume routes.
- Exit Closed C-062 banners should be used for long-duration closures, overlaid across existing G-5 (exit information) signs at a 45-degree angle.
- When planning the temporary closure of an exit ramp, consideration should be given to where traffic will go when it cannot use the exit.
- Alternative exit(s) should be identified to drivers by using dynamic message signs (DMS).

Options:

- A Distance C-130-T tab may be used with the Exit Closed Ahead C-205-A sign if the exit is not visible ahead or there are multiple exits in close proximity.
- For longer duration closures, roadside barriers may be used instead of flexible drums to close the exit.
- For longer duration closures, custom signs or additional dynamic message signs (DMS) providing information on alternative exit(s) may be used.
- Roadside barriers may be offset from the fog line by at least one metre and may include crash attenuation or an appropriate approach flare.
- Where permanent overhead dynamic message signs (DMS) are located in advance of the closure, exit closure information may also be provided on these signs.

Figure 9.15: Temporary Closure of Exit Ramp – Short and Long Duration



9.16 Runaway Lane Open – Short and Long Duration

Purpose:

Runaway lanes are emergency escape ramps created for long, descending grades. On such grades, vehicle speeds may increase and brakes may overheat and fail because of the extensive braking used to slow the vehicle. The runaway lanes allow vehicles with brake problems to exit the roadway and stop. Runaway lanes should be kept open whenever possible because of their importance for roadway safety.

This layout shows the typical setup for work adjacent to a runaway lane that remains open. Effort should be made to **clearly communicate the open status of the runaway lane** to truck drivers and other approaching motorists.

Standard:

- Construction equipment shall never be left parked in—or blocking access to—an open runaway lane.
- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab and a second Right Lane Closed Ahead C-130-R sign shall be placed in advance of the exit ramp.
- A flashing arrow board (FAB) shall be placed inside the taper.
- A buffer space shall always be used.
- A dynamic message sign (DMS) shall be used to communicate the runaway lane status.

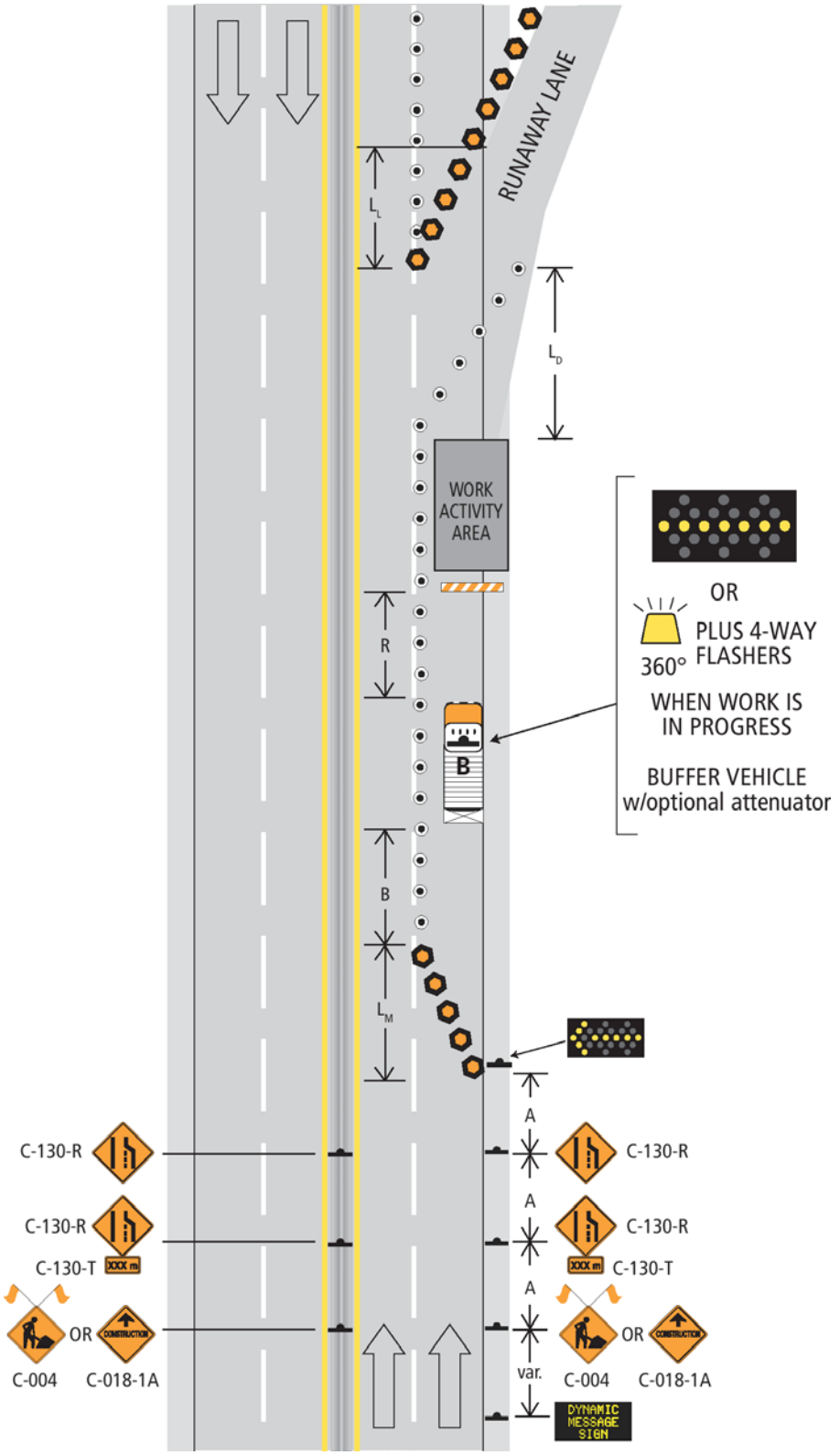
Guidance:

- Signage should be installed along the divided median. If the median space is limited, smaller sized signs may be used.
- Where the speed limit is ≥ 70 km/h a buffer vehicle should be used when workers are present.
- A vehicle-mounted crash attenuator is recommended for use on buffer vehicles.
- Vehicles should not be queued in advance of—or through—the entrance to a runaway lane.

Options:

- None at this time.

Figure 9.16: Runaway Lane Open – Short and Long Duration



9.17 Runaway Lane Closed – Short and Long Duration

Purpose:

Runaway lanes are emergency escape ramps created for long, descending grades. On such grades, vehicle speeds may increase and brakes may overheat and fail because of the extensive braking used to slow the vehicle. The runaway lanes allow vehicles with brake problems to exit the roadway and stop. Runaway lanes should be kept open whenever possible because of their importance for roadway safety.

This layout shows the typical setup for work in front of a runaway lane that is closed. When work is occurring in front of runaway lanes, effort shall be made to clearly communicate the closed status of the runaway lane to truck drivers and other approaching motorists.

Even if a runaway lane is closed, extreme caution should be used when working in front of or across the runaway lane. Never assume that just because it is considered closed, a truck would not attempt to use it in an emergency.

Standard:

- Truck drivers shall be notified of runaway lane closures at (within or in advance of) the brake check and in advance of the grade.
- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab and a second Right Lane Closed Ahead C-130-R sign shall be placed in advance of the entrance ramp.
- The Runaway Lane Closed C-067 sign shall be positioned 200 metres or distance A, whichever is greater, in advance of the affected runaway lane. It shall be removed or covered as soon as possible once the runaway lane is available.
- For construction zones across runaway lanes, use the Closed C-061 tab mounted above or across the existing runaway signs.
- A dynamic message sign (DMS) shall be used to communicate the runaway lane closure. Appropriate positioning and messaging should be discussed with the Road Authority.
- A buffer space shall always be used.

Guidance:

- Signage should be installed along the divided median. If the median space is limited, smaller sized signs may be used.
- Where the speed limit is ≥ 70 km/h a buffer vehicle should be used when workers are present.
- A vehicle-mounted crash attenuator is recommended for use on buffer vehicles.
- Vehicles should not be queued in advance of—or through—the entrance to a runaway lane.
- If a lane closure is not required, the Right Lane Closed Ahead C-130 series and flashing arrow board (FAB) can be removed.

Options:

- Notification at the brake check may be communicated through a dynamic message sign (DMS), C-067 Runaway Lane Closed sign, or custom signing.